

Database Technologies – NoSQL Assignment

1. Create a new database named CDAC and connect to it.
2. Display the list of collections in the CDAC database.
3. Create a collection named LIBRARY.
4. Insert a document into LIBRARY with: title: MongoDB programming, author: Amit, likes: 100
5. Display the inserted document and note the `_id` field.

```
[admin> use CDAC
switched to db CDAC
[CDAC> show collections

[CDAC> db.LIBRARY.insertOne({title: "MongoDB programming", author: "Amit", likes: 100})
{
  acknowledged: true,
  insertedId: ObjectId('69b966125bda97d5fb0fee09')
}
[CDAC> db.LIBRARY.find()
[
  {
    _id: ObjectId('69b966125bda97d5fb0fee09'),
    title: 'MongoDB programming',
    author: 'Amit',
    likes: 100
  }
]
CDAC> █
```

6. Insert another document into LIBRARY with:

title: MySQL programming, authors: [Jack, Jill], likes: 200

7. Display all documents in the LIBRARY collection.

```
[CDAC> db.LIBRARY.insertOne({title: "MySQL programming", authors: ["Jack", "Jill"], likes: 200})
{
  acknowledged: true,
  insertedId: ObjectId('69b966665bda97d5fb0fee0a')
}
[CDAC> db.LIBRARY.find()
[
  {
    _id: ObjectId('69b966125bda97d5fb0fee09'),
    title: 'MongoDB programming',
    author: 'Amit',
    likes: 100
  },
  {
    _id: ObjectId('69b966665bda97d5fb0fee0a'),
    title: 'MySQL programming',
    authors: [ 'Jack', 'Jill' ],
    likes: 200
  }
]
[CDAC> ]
```

8. Display only the first inserted document.

```
[CDAC> db.LIBRARY.find().sort().limit(1)
[
  {
    _id: ObjectId('69b966125bda97d5fb0fee09'),
    title: 'MongoDB programming',
    author: 'Amit',
    likes: 100
  }
]
[CDAC> db.LIBRARY.find({_id:ObjectId('69b958ab5bda97d5fb0fee07')})
[]
[CDAC> db.LIBRARY.find({_id: ObjectId('69b966125bda97d5fb0fee09')})
[
  {
    _id: ObjectId('69b966125bda97d5fb0fee09'),
    title: 'MongoDB programming',
    author: 'Amit',
    likes: 100
  }
]
[CDAC> ]
```

9. Display documents using the pretty() method.

```
[CDAC> db.LIBRARY.find().pretty()
[
  {
    _id: ObjectId('69b966125bda97d5fb0fee09'),
    title: 'MongoDB programming',
    author: 'Amit',
    likes: 100
  },
  {
    _id: ObjectId('69b966665bda97d5fb0fee0a'),
    title: 'MySQL programming',
    authors: [ 'Jack', 'Jill' ],
    likes: 200
  }
]
```

10. Update the document where author = Amit to Amit Sharma.

11. Delete all documents with likes = 100.

```
[CDAC> db.LIBRARY.updateOne({author: 'Amit'}, {$set: {author: 'Amit Sharma'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
[CDAC> db.LIBRARY.deleteOne({likes: 100})
{ acknowledged: true, deletedCount: 1 }
[CDAC> db.LIBRARY.find()
[
  {
    _id: ObjectId('69b966665bda97d5fb0fee0a'),
    title: 'MySQL programming',
    authors: [ 'Jack', 'Jill' ],
    likes: 200
  }
]
```

12. Drop the LIBRARY collection.

13. Drop the CDAC database.

```
[CDAC> db.LIBRARY.drop()
true
[CDAC> show collections

[CDAC> db.dropDatabase('CDAC')
{ ok: 1, dropped: 'CDAC' }
[CDAC> show databases
admin      40.00 KiB
classwork  80.00 KiB
config     108.00 KiB
local      72.00 KiB
CDAC> █
```